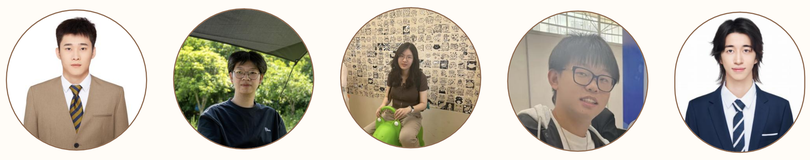
Xi'an Jiaotong-Liverpool University 西交利物浦大学

XJTLU | MUSEUM

Intelligent Interactive Exhibits: Incorporating Augmented Reality and Artificial Intelligence to Enhance the Museum Visit Experience



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1. Introduction and Problem Statement

Museums are vital for preserving cultural heritage and spreading knowledge, they offer a view into the past, exposing visitors to diverse civilizations. However, traditional museum experiences often can't fully engage modern audiences [1]. Visitors today face challenges like information overload [1], lack of personalized guides [1], difficulty connecting with exhibits [2], and limited social interaction [2, 3]. It has been shown in previous study [4] that Large Language Models (LLMs) hold great promise in cultural heritage. They can offer personalized tours and boost interactivity [4]. By focusing on human needs, behaviors, and emotions and integrating Augmented Reality (AR) and Artificial Intelligence (AI), this project turns passive museum visits into engaging, interactive, personalized journeys, significantly improves the immersive visiting experience. Our study aims to answer: **RQ1:** ***Can this museum intelligent interactive exhibit system enhance the museum visit experience?*** **RQ2:** ***If so, to what extent can it enhance the visit experience.***

2. Design and Methodology

Discovering Requirements	• **Stakeholder interview** (5 Visitors and 1 Museum Professional) • **Persona & Context** • **Competitive product analysis**	• **High information load** (N=4) • **Not personalized** (N=4) • **Difficulty connecting with exhibits** (N=2) • **Want to connect with other visitors** (N=2) • **Off-site exhibits** (N=2)	• **Brainstorm:** AR Intro, Chat, Posts, Link, AI Map, Share with friends, Achievement System, AR Ticket, etc.	• **Reduce to four functions:** AR Intro, Chat, Posts, Link.
Designing Alternatives	• **Wireframe** (Low-fidelity prototypes)	• **Several button layout designs** • **Several chatting interface designs** • **Several link exhibits interface designs**	• **A/B testing** • **Questionnaire** • **Stakeholder interview**	• **Determine the best UI solution** **Redesign based on the feedback**
Prototyping	• **High-fidelity prototypes**	• **Heuristic evaluation**	• **Stakeholder Test & Interview**	**Redesign based on the feedback**
Evaluation	• **Pilot study** • **On-site user test at XJTLU Museum**	**Quantitative analysis:** • Museum experience • Multimedia guide experience • System usability • User experience	**Qualitative analysis:** • User interview	**Redesign based on the feedback**

Figure 1: Design Process.

3. Implementation and Outcomes

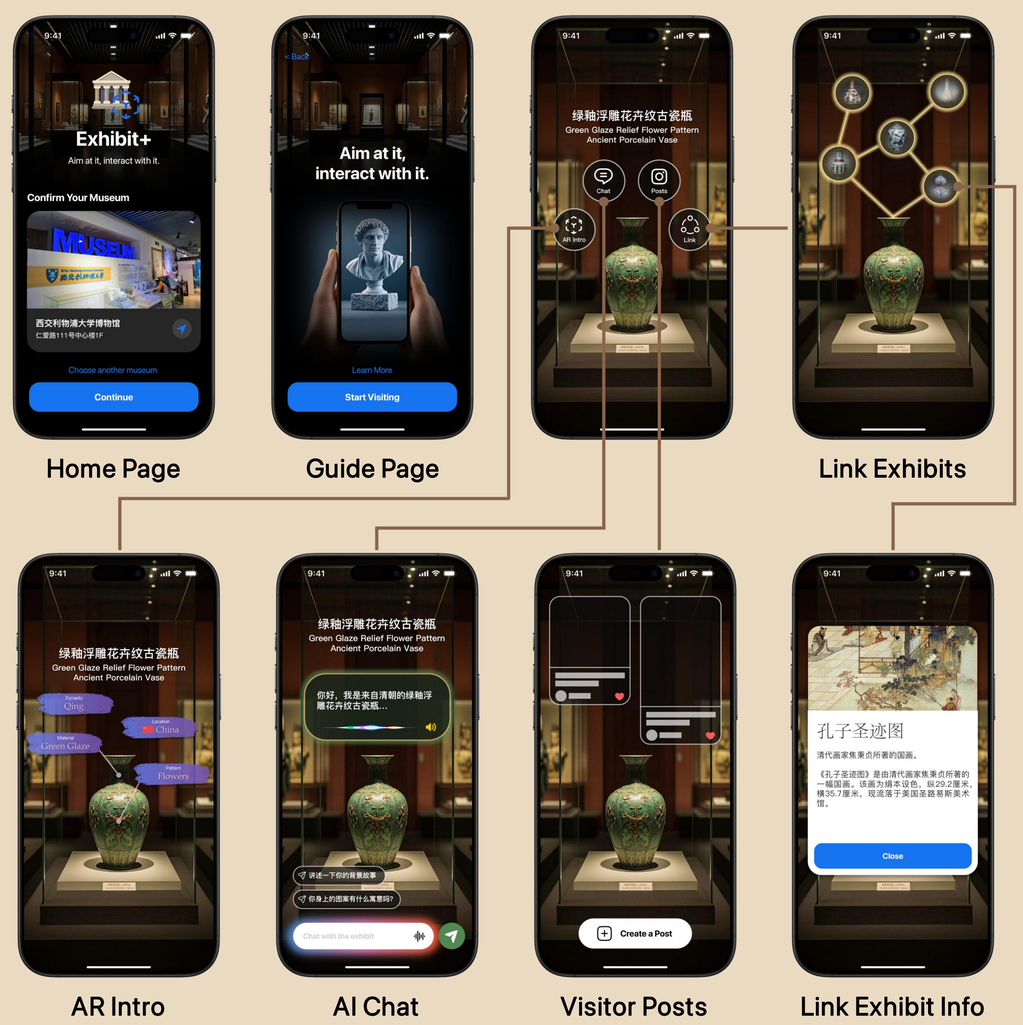

Figure 2: High-fidelity Prototype of Intelligent Interactive Exhibit System.

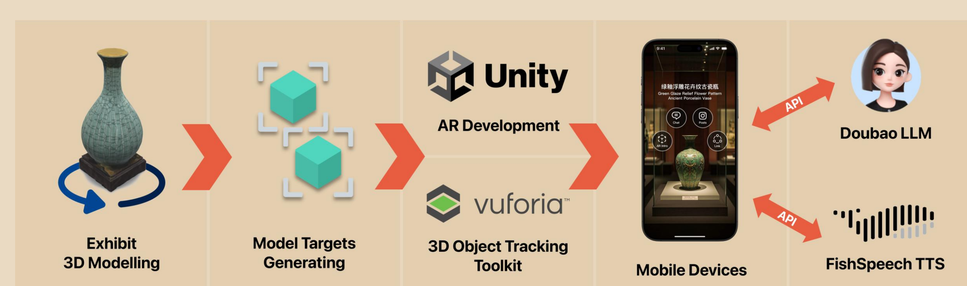

Figure 3: System Implementation. AR=Augmented Reality, LLM=Large Language Model, TTS=Text-to-Speech Model.

4. Evaluation and Impact

We invited 14 (N=14) individuals of different ages (20~22) and backgrounds to the XJTLU Museum for an on-site user test. We set up 6 exhibits and participants visited them with and without the IIE system. The final System Usability Score is 72.86 (Good Usability).

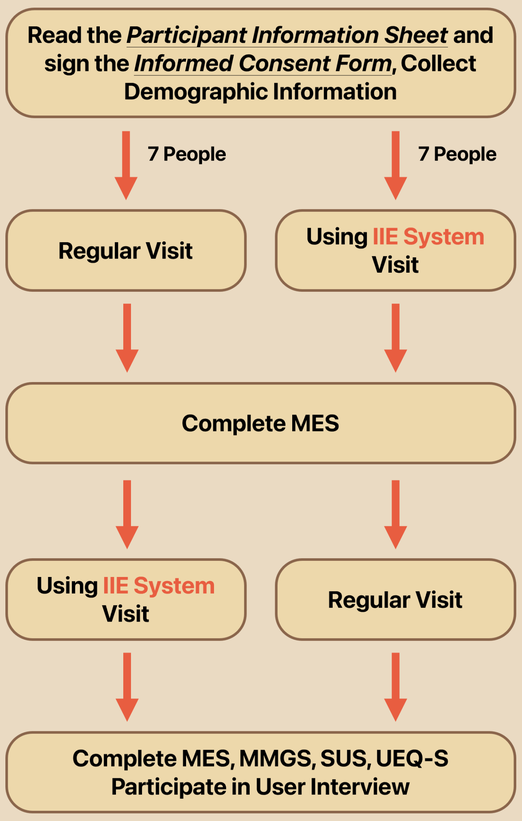

Figure 4: User test flow chart. IIE: Intelligent Interactive Exhibit, MES: Museum Experience Scale, MMGS: Multimedia Guide Scale, SUS: System Usability Scale, UEQ-S: Short Version of the User Experience Questionnaire.

Gender	Male		Female		Non-binary
	8		6		0
Frequency of museum visits	Never	Once every few years	1-2 times per year	3-4 times per year	5 times or more per year
	0	4	9	0	1
Types of museums they like to visit	Art Museums	History Museums	Science Museums	Other	
	5	4	4	1	
Technology Familiarity	Never heard of	Heard of, but not understood	Some understanding	Familiar	Very familiar
LLM	0	2	5	5	2
AR	0	6	6	2	0
TTS	7	5	2	0	0

Figure 5: Demographic Information.

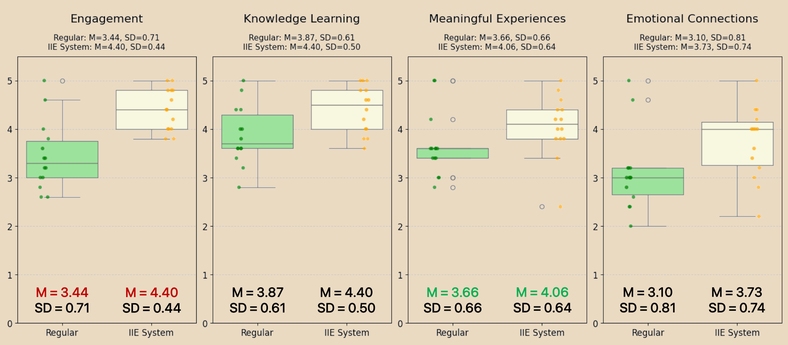

Figure 6: Results of Four Dimensions in MES (Regular Visit and Using IIE System Visit).

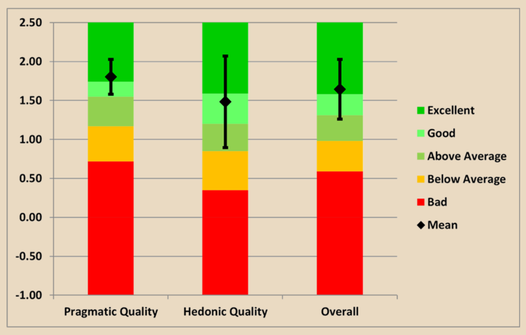

Figure 8: UEQ-S Result.

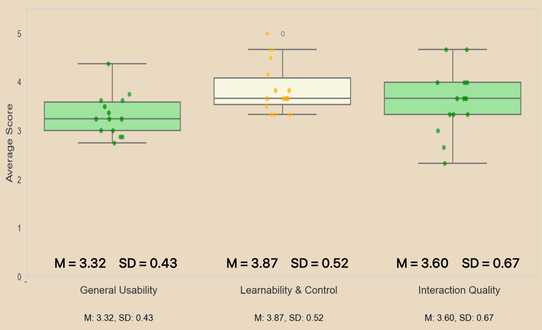

Figure 7: Result of Three Dimensions in MMGS (Using IIE System Visit).

Interview Result

- Answers generated by AI can be more detailed (N=5).
- A tone that is more in line with the temperament of exhibits (N=4).
- The interactive panel should always face the user (N=3).
- More interactive animations (N=2).

5. Conclusion and Future Work

Results show that this intelligent interactive exhibit system significantly improves the museum visit experience. It also had good usability and excellent user experience, which indicated that design of this system is successful and fully embodies the human-centered design principle.

Limitations

- Due to the limitations of 3D model tracking technology, reflective, transparent, glossy, smooth, or translucent exhibits may not be accurately recognized.
- The Text-to-Speech (TTS) model requires some time to generate the synthesized voice.

Future Work

- Explore more efficient object tracking solutions to avoid the process of 3D scanning and modeling for each exhibit.
- Explore the feasibility of extending this system to larger museums or exhibitions, such as historical artifact museums and art exhibitions.

References

[1] J. H. Falk and L. D. Dierking, *The Museum Experience Revisited*, 0 ed. Routledge, 2016. doi: 10.4324/9781315417851.

[2] J. Roberts, A. Banerjee, A. Hong, S. McGee, M. Horn, and M. Matcuk, "Digital Exhibit Labels in Museums: Promoting Visitor Engagement with Cultural Artifacts," in *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, Montreal QC Canada: ACM, Apr. 2018, pp. 1–12. doi: 10.1145/3173574.3174197.

[3] N. Simon, *The participatory museum*. Santa Cruz, California: Museum 2.0, 2010.

[4] J. Liang, G. Zeng, Y. Li, and Y. Dong, "ARTimeTravel: Understanding Spatial Changes in Heritage Sites Over Time through Web-Based Augmented Reality Serious Games," *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, Yokohama, Japan, Apr. 26–May 1, 2025, pp. 1–8. Available: https://doi.org/10.1145/3706599.3719904.